Optimizing Control Center Placement to Enhance the Robustness of Cyber-Physical Power Systems: A Complex Network Perspective

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(Dated: 16 April 2026)

As a typical interdependent critical infrastructure network, cyber–physical power system (CPPS) is increasingly vulnerable to cascading failures, making robustness enhancement a problem of paramount importance. This paper proposes a novel analysis framework and optimization method for CPPS robustness that explicitly considers distributed control centers in the communication layer. Specifically, we first extend the centralized control architecture to a distributed paradigm, and introduce a Functional Viability-Constrained Island Detection mechanism, which imposes stricter connectivity and functional constraints on island viability in both the power and communication layers. Then, we systematically investigate how this deployment influences the robustness of CPPS against cascading failures under different coupling patterns. Finally, under limited control resources, we develop a Cohesion-Propagation Importance Metric that integrates multiple centrality features to identify optimal control center locations, effectively balancing local structural stability with global functional reach to enhance overall system robustness. Extensive experiments on benchmark systems under various coupling scenarios demonstrate that our method significantly enhances the robustness of CPPS, offering a practical and cost-effective solution for strengthening CPPS robustness.

I. INTRODUCTION

The deep integration of physical infrastructure with communication networks, propelled by advances in control and computing technologies, has catalyzed the evolution of traditional power grids into Cyber-Physical Power Systems (CPPS)[1]. This integration, while yielding significant improvements in observability and operational efficiency, concurrently broadens the attack surface. Consequently, the rising incidence of cyber attacks and their potential for severe socio-economic disruption necessitate a rigorous investigation into the resilience and stability of these interdependent critical infrastructures. Addressing this challenge is therefore of paramount importance for ensuring the secure and reliable operation of modern energy systems.

As a typical interdependent network, CPPS have been extensively studied from the perspective of robustness, with research generally spanning three interrelated dimensions: system modeling, cascading failure dynamics, and robustness enhancement strategies. In terms of modelling, complex network theory is widely utilized to represent the CPPS as a coupled network. Specifically, the communication layer is frequently modelled as a scale-free network, often implementing a centralized control mode with a designated control center[2,3]. Additionally, scale-free communication topologies incorporating community structures have been constructed to reflect the inherent properties of power systems. Regarding cyber threats, studies have investigated Advanced Persistent Threats, noting their higher potential for damage compared to standard Denial-of-Service attacks[4] or false data injection attacks[5–7]. Finally, considering that inter-network coupling facilitates fault propagation across subsystems, the impact of interdependence strength has been extensively analyzed. These investigations cover various coupling schemes, ranging from strong[8] and partial[9] to weak interdependencies[10]. The cascading failure processes in CPPS can be broadly categorized into three theoretical frameworks[11]. The first employs percolation theory, spanning general analytical formulations[12] to dynamic models based on multilevel percolation systems[13]. The second utilizes Markov chains to capture stochastic transition processes, including scenarios involving multi-state component behaviors[14,15]. The third encompasses probabilistic or deterministic simulations that incorporate physical power flow constraints. Within this domain, DC power flow models have been applied to assess resilience against random attacks and to simulate cascades under shortest-path routing constraints[16]. More sophisticated simulations employ AC power flow to investigate topological attacks on buses and branches[17], or combine DC and AC optimal power flow to account for diverse attack vectors and node connectivity mechanisms[18]. Recently, advances in artificial intelligence have also facilitated the application of deep learning methods to the modeling of cascading dynamics[19].

Within the dimension of robustness enhancement strategies, identifying and protecting critical components constitutes a fundamental approach to mitigating cascading failures in CPPS. Complex network theory has proven particularly effective for formulating such optimization strategies. For instance, critical nodes have been identified using fusion degree-centered metrics, while comprehensive protection schemes integrate multiple indicators including degree, betweenness, and performance metrics. The evolution of node importance evaluation follows a clear trajectory. Early studies relied on fundamental structural characteristics such as degree centrality[20], betweenness centrality[21], and k-shell decomposition[22]. While foundational, these single-dimension metrics often lack universality across diverse network topologies. To address this, researchers refined these basic metrics, developing algorithms like enhanced shell-core

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centrality[23] and cumulative shell models[22]. However, approaches relying on a single topological feature remain inherently limited. Consequently, recent methodologies have shifted towards integrating multiple structural functionalities, simultaneously considering factors such as k-shell values and second-order neighborhoods[24], or synthesizing node connectivity with neighboring node influence. To further capture higher-order interactions, scholars have incorporated physical principles—most notably gravitational formulas—to construct gravity-based models for identifying influential nodes[25,26]. These models have been subsequently enhanced by incorporating effective distance and reverse random walks[27], as well as node layout considerations[28]. Compared to traditional topological measures, these physics-inspired metrics yield results with greater physical significance regarding propagation influence[25]. Despite these advances, the application of such methods to coupled interdependent networks like CPPS remains underexplored, with most existing studies validated on isolated single-layer networks.

Despite these advances, research on CPPS robustness remains in its early stages, with several critical issues warranting further investigation. First, as power systems continue to scale up, the traditional centralized control model for the communication layer has become increasingly inadequate[31]. Second, while the mechanism of full-type node connectivity survival holds significant practical relevance for analyzing cascading failures on the power system side, confining this mechanism to a single subsystem is insufficient given that CPPS constitutes a coupled system where the communication layer is equally vulnerable to cyberattacks. Third, some studies that analyze catastrophic cascading failures fail to propose concrete robustness enhancement strategies[29,32]. Moreover, even when effective node identification strategies based on complex network theory have been developed, these are often validated solely on isolated real-world networks—such as NetScience[33] or Power[34]—rather than on coupled interdependent systems like CPPS. Whether such optimization strategies retain their effectiveness for interdependent networks thus remains an open question. To contextualize the present work, Table 1 summarizes the key distinctions between this study and the relevant literature. Specifically, the contributions of this paper are threefold:

- **Distributed control architecture with Functional Viability-Constrained Island Detection (FVC-ID) mechanism.** This paper adopts a distributed control architecture featuring multiple control centers, and introduces a Functional Viability-Constrained Island Detection (FVC-ID) mechanism that imposes stricter connectivity and functional constraints on island viability in both layers, addressing the critical limitation of conventional island detection that overlooks closed-loop operational requirements.

- **Systematic analysis of control center deployment under diverse coupling patterns.** Building on this architecture, we systematically investigate how the quantity and distribution of control centers influence CPPS robustness against cascading failures across various coupling scenarios, revealing the nuanced relationship between control redundancy and system resilience.

- **Cohesion-Propagation Importance Metric-based optimal placement of control centers under resource constraints.** Recognizing inherent resource limits, we develop a Cohesion-Propagation Importance Metric (CPIM) that integrates multiple centrality features to identify optimal control center locations, balancing local structural stability with global functional reach to enhance robustness without substantial additional costs—achieving the dual objectives of mitigating cyber attacks and strengthening system robustness.

The remainder of this paper is as follows. Section 2 presents the models of power system and communication networks, along with the coupling mechanisms between two layers. Section 3 introduces the cascading failure process model and the FVC-ID mechanism. Simulation results are presented in Section 4, and conclusions are summarized in Section 5.

II. CPPS MODEL

This section presents the topological modeling of Cyber-Physical Power Systems under distributed control architectures. As illustrated in Fig. 1, a CPPS comprises two interdependent subsystems: the power system and the communication system. We describe the structural representation of each subsystem and their coupling mechanisms, laying the foundation for subsequent robustness analysis.

A. Power network model

The power network serves as the physical foundation of a cyber-physical power system, responsible for generating, transmitting, and distributing electrical energy to meet load demands. Its reliable operation is monitored and controlled via the coupled communication network. The power network can be represented as a directed graph $G_P = (V_P, E_P)$, where V_P denotes the set of power nodes (buses) and E_P represents the transmission lines connecting them. Consider a power system consisting of n nodes and l transmission lines. The active power injected at each node is given by the column vector $P = [p_1, p_2, \ldots, p_n]^T$.

When a fault occurs in the power grid, the affected buses are disconnected, altering the network topology and triggering power flow redistribution. To capture such dynamics, both direct current (DC) and alternating current (AC) power flow models are widely adopted. Compared to its AC counterpart, the DC power flow model offers a linearized approximation of system behavior, significantly reducing computational complexity while retaining essential network characteristics. Accordingly, this study adopts the DC power

TABLE I. Comparison of this study with related literature from five perspectives.

	Power-layer Operability	Cyber-layer Operability (FVC-ID)	Control Center Nodes	Distributed Control	Robustness Optimization
Our work	✓	✓	✓	✓	✓
18	✓	×	✓	×	✓
10	×	×	✓	✓	✓
53	×	×	✓	×	✓
29415	×	×	✓	✓	×
1308	×	×	✓	×	×
111912	×	×	×	×	×

✓: Mechanism or strategy is explicitly considered, ×: Not considered

flow framework for power system analysis, in which the active power injections are computed as follows:

$$P = B\Theta \tag{1}$$

where $\Theta = [\theta_1, \theta_2, \theta_3, \ldots, \theta_n]^T$ denotes the phasor column vector of the power node, $p_i > 0$ represents a source node such as a generator, and $p_i < 0$ denotes a load node such as a consumer. In this context, B is a node admittance matrix, which is an $n \times n$ matrix defined as:

$$B = \begin{bmatrix} \sum_{j\neq 1} b_{1j} & -b_{12} & -b_{13} & \cdots & -b_{1n} \\ -b_{21} & \sum_{j\neq 2} b_{2j} & -b_{23} & \cdots & -b_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -b_{n1} & -b_{n2} & -b_{n3} & \cdots & \sum_{j\neq n} b_{nj} \end{bmatrix} \tag{2}$$

where b_{ij} is defined as follows:

$$b_{ij} = b_{ji} = \begin{cases} \frac{1}{x_{ij}}, & \text{if nodes } i,\, j \text{ are connected} \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where x_{ij} is the admittance of transmission line between nodes i and j. The power flow can be computed as:

$$Z = Y^{-1} A^T \Theta \tag{4}$$

where $Z = [z_1, z_2, z_3, \cdots, z_e]^T$ represents the active power flow within the transmission line. $Y = diag(y_e)$ is the diagonal matrix of where y_e denotes the admittance of transmission line e. A is the line-node association matrix. Its entry a_{ij} indicates the incidence of line i with node j: $a_{ij} = 1$ if j is the source node, $a_{ij} = -1$ if j is the target node, and $a_{ij} = 0$ otherwise.

B. Communication network model

As illustrated in Fig. 1, the communication layer is modeled as an undirected graph $G_C = (V_C, E_C)$, where V_C and E_C denote the node and edge sets, respectively. The nodes consist of two functional types: intelligent electronic devices (IEDs) and control centers. IEDs—comprising sensors, routers, and actuators—monitor the grid, process control signals, and facilitate inter-node communication. Control centers aggregate data from IEDs to generate coordinated control decisions. Given their operational criticality[5], control centers are assumed to be secure and resilient to disruptions. Unlike traditional centralized architectures that rely on a single, isolated control centers[18], this work adopts a distributed model with multiple control centers deployed within the communication layer, none of which are directly coupled to power nodes. Owing to the scale-free, heavy-tailed degree distribution typical of communication networks, the topology is generated using the Barabási–Albert (BA) model. Construction starts with $m_0 = 4$ fully connected nodes; each new node then attaches via $m = 2$ preferential links, yielding a scale-free network. In most real-world communication networks, information flows along the shortest paths between nodes. Without loss of generality, this paper employs betweenness centrality to quantify the structural load value and information-carrying capacity of nodes and edges in the communication layer, defined as follows:

$$\begin{cases} B_s = \sum_{i\neq j} \frac{\delta_{ij}(s)}{\delta_{ij}}, \\ B_t = \sum_{i\neq j} \frac{\delta_{ij}(t)}{\delta_{ij}}. \end{cases} \tag{5}$$

where δ_{ij} denotes the total number of shortest paths between nodes i and j, and $\delta_{ij}(s)$ and $\delta_{ij}(t)$ represent the number of such paths passing through node s and edge t, respectively.

C. Coupling patterns of CPPS and interdependency mechanisms

The CPPS is formed through interdependent coupling between the power and communication systems, as depicted by the dashed channels in Fig. 1. Power nodes supply

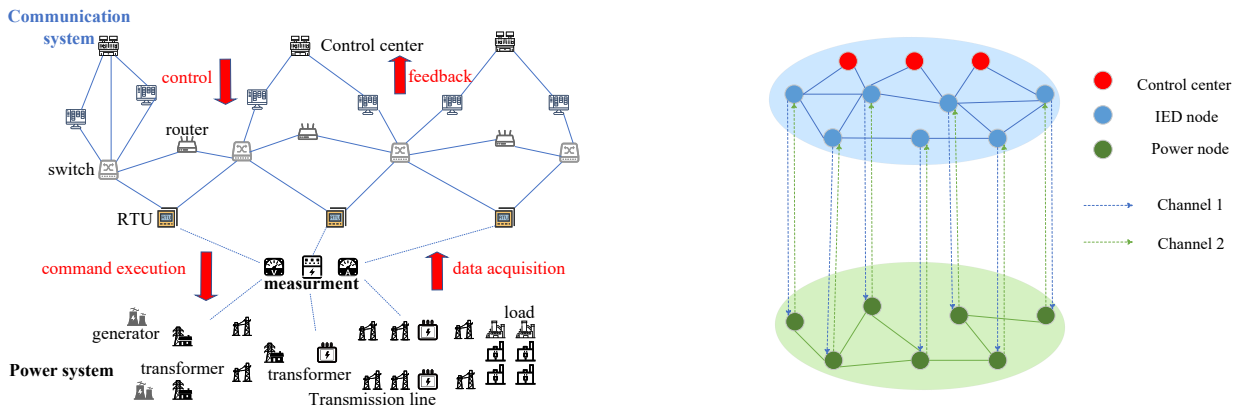


Fig. 1. The Schematic illustration of CPPS with distributed control centers.

energy to IED nodes (Channel 1), while IED nodes transmit monitoring and control data to power nodes (Channel 2). To ensure complete observability and controllability of the power network, a one-to-one coupling scheme is adopted between power nodes and IEDs. Control centers manage all IED nodes globally and are equipped with independent backup power supplies, thus remaining decoupled from the power layer.

Three distinct one-to-one coupling strategies are examined to evaluate their impact on system robustness:

Coupling Mode I – Betweenness Assortative Coupling (BAC): Nodes in both networks are sorted by descending betweenness and coupled sequentially, pairing high-betweenness power nodes with high-betweenness communication nodes.

Coupling Mode II – Betweenness Disassortative Coupling (BDC): Power nodes are sorted by ascending betweenness while communication nodes are sorted by descending betweenness, creating cross-scale coupling between low-betweenness power nodes and high-betweenness communication nodes.

Coupling Mode III – Random Connection (RC): Power and communication nodes are coupled randomly without topological constraints.

While the aforementioned strategies define the static topological connections between the two networks, the actual interactions during a disruptive event exhibit stochastic characteristics. In interdependent networks, node interactions are classified as either strong or weak interdependency. Under strong interdependency, a node failure in one network deterministically triggers the failure of its coupled counterpart. However, given the distinct operational characteristics and buffering capabilities of power and communication subsystems in CPPS, this paper adopts the weak interdependency framework to model cross-layer failure propagation. Specifically, when a node fails, its coupled counterpart in the dependent network fails with a probability p_{fail}, where $0 \leq p_{\text{fail}} \leq 1$. This probabilistic formulation captures the stochastic nature of cyber-physical interactions more realistically than rigid deterministic models, thereby accurately reflecting the fault propagation threshold across layers.

D. Functional Viability-Constrained Island Detection (FVC-ID) mechanism

The survival of a system during cascading failures cannot be guaranteed by a robust static topology alone. Equally critical—yet often overlooked in conventional approaches—are the criteria used to determine whether isolated components remain operational. Traditional island detection algorithms adopt an overly optimistic assumption, classifying any connected Island containing a generator as an operational power island. This purely structural perspective, however, fundamentally disregards a basic physical law of power systems: maintaining a closed-loop power flow strictly requires the simultaneous presence of both generation and load.

To address this fundamental gap, this paper introduces the Functional Viability-Constrained Island Detection (FVC-ID) mechanism—a novel framework designed to rigorously evaluate the true survivability of Island across both layers. In the power layer, the FVC-ID criterion stipulates that a connected component is considered functional only if it contains at least one generator node and one load node, thereby enforcing the essential closed-loop condition overlooked by structural approaches. Extending this principle to the communication layer under a distributed architecture, an island is deemed operational only if it can sustain closed-loop control functionality—either through the co-location of at least one IED and one control center, or through the presence of autonomous control centers alone.

By explicitly filtering out structurally connected yet functionally paralyzed components—termed ”zombie islands” —this dual-layer mechanism bridges the critical gap between graph-theoretic connectivity and engineering reality. More importantly, it prevents the severe overestimation of CPPS robustness that plagues traditional models, ensuring that assessments are grounded in actual operational feasibility rather than idealized topological assumptions.

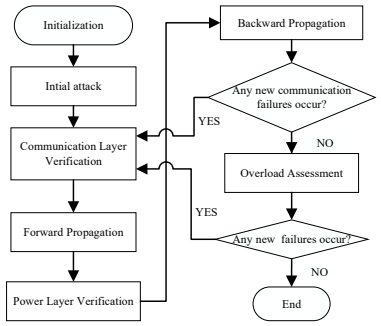


Fig. 2. The flowchart of cascading failure process in CPPS.

III. CASCADING FAILURE MODEL

The transition from localized cyber disruptions to widespread system collapse is governed by complex propagation dynamics. With the underlying topological models and operational constraints established, this section shifts focus to the dynamic evolution of cascading failures. First, the iterative progression of faults—from the initial cyber attack to cross-layer physical breakdowns—is systematically modeled. While understanding this collapse mechanism is crucial, proactively preventing it represents the ultimate goal of resilience analysis. Therefore, rather than relying on costly physical capacity redundancy, this section subsequently introduces a novel network metric. This specific metric is designed to strategically optimize the deployment of control centers, effectively thwarting catastrophic failure propagation at its structural roots.

A. Cascading failure process

Integrating the weak interdependency model and the FVC-ID mechanism, the cascading failure dynamics unfold through the following iterative procedure, as illustrated in Fig. 2:

Step 1: Initialization: Initialize the coupled CPPS topology. Configure power system parameters in Matpower and compute the initial DC power flow distribution.

Step 2: Initial Attack: Remove a target node from the communication layer to trigger the cascade.

Step 3: Communication Layer Verification: Apply the FVC-ID criterion to all connected components in the communication layer. Mark any component that violates the all-type connectivity condition as failed.

Step 4: Forward Propagation: For each failed IED node, its coupled power node fails with probability p_{fail} according to the weak interdependency model.

Step 5: Power Layer Verification: Apply the FVC-ID criterion to the power layer. Remove any island that lacks both generators and loads.

Step 6: Backward Propagation: For each newly failed power node, its coupled communication node fails probabilistically. If new communication failures occur, return to Step 3; otherwise, proceed to Step 7.

Step 7: Overload Assessment: The system checks for capacity violations in both layers. In the communication layer, overloaded nodes or links are identified and removed. In the power layer, the DC power flow is recomputed, and transmission lines exceeding their thermal capacity constraints are eliminated. If this step induces new failures, return to Step 3; otherwise, the cascading failure terminates and the final network state is recorded.

When the iterative process eventually halts with no further overload or isolation events, the network reaches an equilibrium state. To quantitatively evaluate the damage degree and resilience of the CPPS at this final state, the paper define the Node Survival Rate (NSR) as the primary evaluation metric:

$$NSR = \frac{N_{sur}}{N} \tag{6}$$

where N_{sur} represents the number of nodes surviving in CPPS, and N represents the total number of nodes in the initial CPPS.

B. The Cohesion-Propagation Importance Metric

The cascading failure dynamics described above highlight the acute vulnerability of CPPS under targeted attacks. To mitigate such catastrophic propagation without resorting to brute-force redundant expansions, strategically deploying control centers at the most structurally critical locations is paramount. Therefore, this section introduces the foundational centrality measures and our novel optimization strategy. Bae et al.[22] developed the neighborhood coreness centrality (C_{nc+}), which aggregates the K-shell values of a node's neighbors:

$$\begin{cases} C_{nc}(i) = \sum\limits_{j \in \psi(i)} k_s(i), \\ C_{nc+}(i) = \sum\limits_{j \in \psi(i)} C_{nc}(i). \end{cases} \tag{7}$$

where $\psi(i)$ represents the set of adjacent neighbors of i, $k_s(i)$ is the K-shell value at a certain node within this neighborhood. Drawing inspiration from the law of universal gravitation, Ma et al.[25] proposed the extended gravity centrality (G_+) to quantify a node's propagation influence and is formulated as:

$$\begin{cases} G(i) = \sum\limits_{j \in \psi(i), j \neq i} \dfrac{k_s(i) k_s(j)}{d(i,j)^2}, \\ G_+(i) = \sum\limits_{j \in \psi(i)} G(j). \end{cases} \tag{8}$$

where $d(i,j)$ is the shortest distance between nodes i and j.

The theoretical elegance of these established metrics notwithstanding, their direct application to CPPS exposes critical limitations. Most traditional centrality measures—and even recent composite algorithms—are fundamentally optimized for isolated, single-layer networks. They operate under

the implicit assumption that basic structural connectivity equates to system survivability. In stark contrast, CPPS constitutes a deeply coupled dual-layer system governed by strict operational constraints. Within this context, a node's criticality is defined not by its abstract topological position, but by its concrete ability to suppress rapid cross-layer cascading avalanches and sustain autonomous operations under the rigorous FVC-ID criteria. Consequently, directly transplanting conventional graph-theoretic metrics into CPPS often results in suboptimal allocation of control resources, ultimately failing to mitigate realistic cyber-physical disruptions.

Individual centrality metrics, each capturing specific topological features, prove inherently insufficient for ensuring the robust operation of a CPPS subjected to complex cascading failures. Consider, for example, prioritizing node protection based solely on degree centrality d. Although this approach may effectively identify locally connected hub nodes, it overlooks their position within the global power flow, potentially leaving critical long-distance transmission bottlenecks unprotected. A different limitation emerges with betweenness centrality B_s: nodes possessing high betweenness are undeniably vital for maintaining global connectivity, yet they often reside in sparsely connected regions. Should such a bottleneck node fail, the absence of local topological redundancy would rapidly fragment the network into isolated micro-islands, rendering them incapable of satisfying the FVC-ID mechanism.

To overcome the inherent limitations of these single-feature metrics, this paper proposes a novel Cohesion-Propagation Importance Metric (*CPIM*). The core design philosophy of *CPIM* is to seamlessly integrate local structural stability with global functional reach—two attributes fundamentally required by the FVC-ID mechanism to ensure realistic island operability. The *CPIM* value for node i is formulated as follows:

$$CPIM(i) = \frac{B_s(i) + k_s(i)}{cloc_i} + (d(i) \times C(i)) \tag{9}$$

where $B_s(i)$, $k_s(i)$, $cloc_i$, d_i, and C_i denote the betweenness centrality, K-shell value, closeness centrality, degree, and clustering coefficient of node i, respectively.

The first term quantifies a node's global propagation potential by integrating K-shell value k_s and betweenness centrality B_s. This combination identifies components that simultaneously occupy the network's hierarchical core and serve as critical routing hubs for cross-layer power and data flows. The inclusion of closeness centrality $cloc_i$ calibrates the metric to reflect global command-and-control efficiency. Operationally, deploying control centers at nodes with high global potential maximizes system observability and accelerates critical signal dispatch during fault initiation.

The second term evaluates local cohesion capacity, which is intrinsically linked to the FVC-ID mechanism. By multiplying degree d with clustering coefficient C, *CPIM* explicitly favors nodes embedded in tightly-knit, redundant local neighborhoods. Physically, nodes with high local cohesion are significantly more likely to form viable, self-sustaining islands during cascading failures. Their dense local interconnections prevent immediate topological isolation of adjacent generators, loads, and IEDs, enabling these cohesive pockets to satisfy FVC-ID criteria and maintain localized autonomous operation even after long-range line tripping.

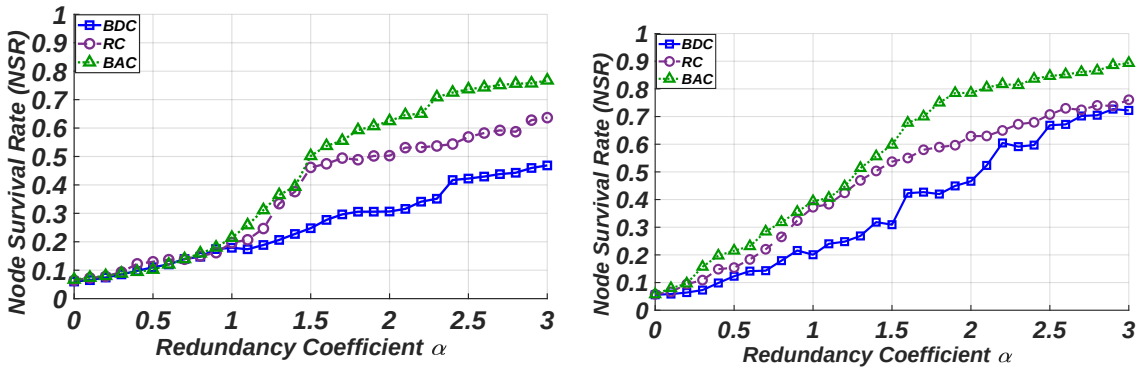
IV. CASE STUDY

Numerical simulations are conducted in this section to empirically validate the proposed theoretical framework and optimization metric. Rather than relying on topological models, the experimental setup constructs realistic CPPS configurations by coupling standard power grids with Barabási-Albert (BA) scale-free communication networks. The simulation focuses on a targeted cyber attack scenario against a communication layer governed by a distributed control architecture. To accurately capture the stochastic characteristics of fault propagation, a probabilistic interdependency model is utilized with the failure propagation probability $p_{\text{fail}} = 0.3$. All the results are averaged over 1,000 independent realizations under the IEEE 39-coupled system and the IEEE 118-coupled system. Through this meticulously designed framework, the system's robustness is comprehensively assessed across three pivotal dimensions: strategies for interdependent coupling, allocation of control resources, and the strategic optimization of node deployment.

A. Robustness Analysis of Distributed Control Centers under Varying Coupling Patterns

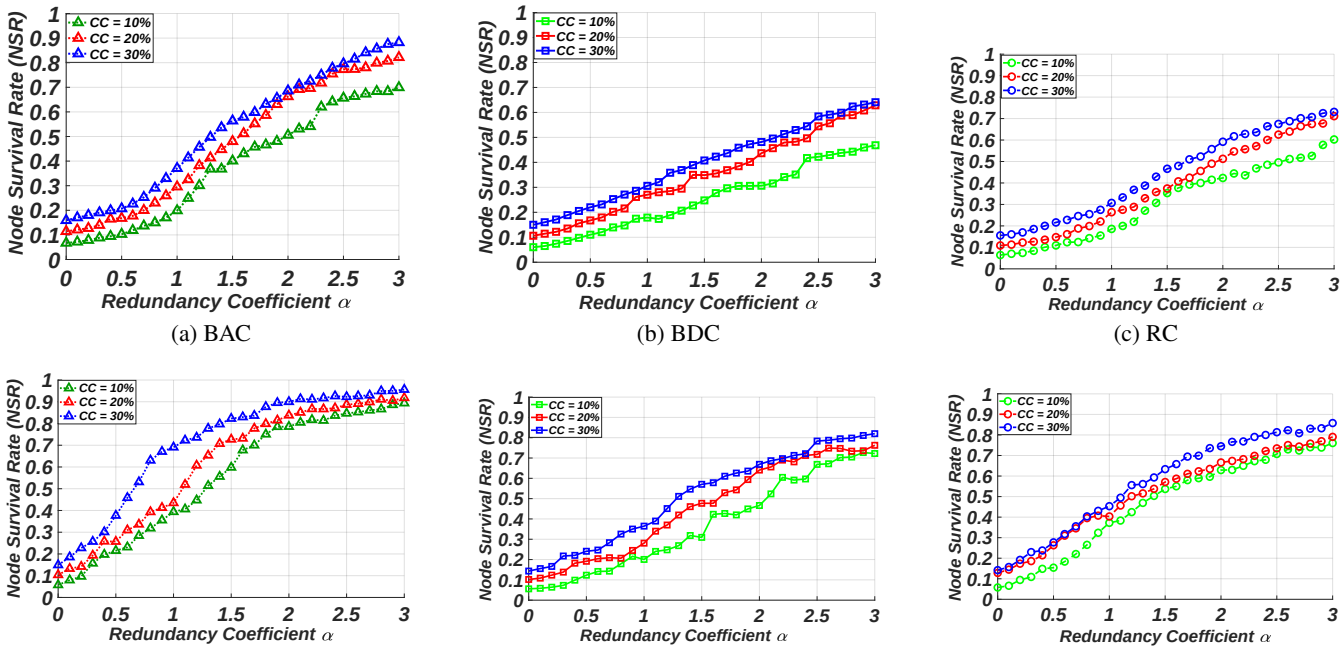
This paper first evaluates the robustness of CPPS under a distributed control architecture, where control centers are colocated with 10% of the power nodes. Fig. 3 presents cascading failure simulation results for the IEEE 39-bus (Fig. 3a) and IEEE 118-bus (Fig. 3b) systems under different coupling patterns.

As shown in Fig. 3(a), when the capacity redundancy coefficient satisfies $\alpha \geq 1$, the betweenness-based assortative coupling (BAC) strategy significantly outperforms other coupling modes. This advantage can be physically interpreted through the proposed functional viability of control-enabled islands with distributed control. Under targeted attacks that fragment the network, BAC ensures that high-betweenness power nodes remain paired with their communication counterparts. The resulting islands are therefore more likely to contain both generation-load pairs and functional control centers, thereby satisfying the FVC-ID criteria. Such configurations prevent the formation of "zombie islands"—fragments that are physically balanced but lack cyber monitoring and control. In contrast, when $\alpha < 1$, the performance gap narrows. This corresponds to a heavily stressed grid operating with limited thermal margins, where physical vulnerabilities dominate. Under such conditions, even a single line outage can trigger cascading thermal



(a) (b)

Fig. 3. The impacts of redundancy coefficient α on CPPS robustness with the distributed control scheme under three coupling patterns in IEEE 39 (a) and IEEE 118 (b) bus systems.



(a) BAC (b) BDC (c) RC

(d) BAC (e) BDC (f) RC

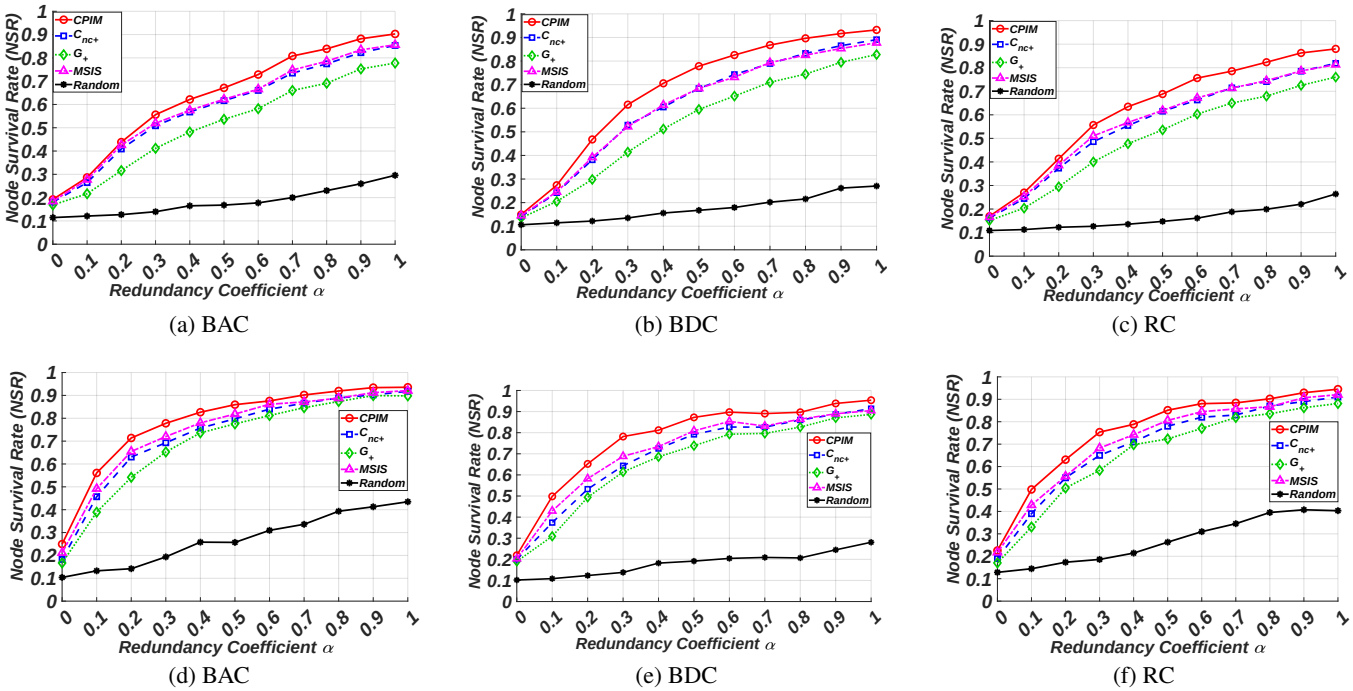
Fig. 4. Robustness Comparison between different proportions of control centers. Figs. 4(a)-(c) and Figs. 4(d)-(f) show the performance of different coupling patterns under IEEE 39-bus system and IEEE 118-bus system.

overloads that evolve too rapidly for cyber intervention to mitigate.

A cross-system comparison between IEEE 39 and IEEE 118 further reveals the scale-dependent nature of structural vulnerability. In the larger IEEE 118 system, performance disparities among coupling strategies become more pronounced. This divergence underscores the necessity of the FVC-ID mechanism in assessing large-scale CPPS. In extensive grids, topological mismatches—such as those in the BDC mode—exhibit a scale-amplification effect. Pairing high-load or high-betweenness power nodes with peripheral communication nodes leads to disjointed interdependency chains. When failures occur, such mismatches generate large, isolated regions that fail FVC-ID requirements, either due to power imbalance or loss of observability. As a result, control failures propagate over longer topological distances. These findings demonstrate that enforcing assortative coupling (BAC) is essential for suppressing long-range cascading failures and ensuring FVC-ID compliance in large-scale systems.

B. Impact of Control Center Proportion on CPPS Robustness

This section further investigates the impact of control center density within a distributed architecture on the robustness of the CPPS. Fig. 4 illustrates the *NSR* across three coupling



(a) BAC (b) BDC (c) RC

(d) BAC (e) BDC (f) RC

Fig. 5. Robustness Comparison between the proposed *CPIM* and four classical placement methods. Figs 5(a)-(c) and (d)-(f) show the results of different coupling patterns under the IEEE 39-bus system and IEEE 118-bus system.

scenarios as the proportion of control centers varies.

The results reveal a distinct positive correlation: a higher proportion of control nodes is generally associated with an improved final survival rate. At the physical layer, increasing the number of distributed control centers reduces the cyber-distance between IEDs and their respective controllers. More importantly, under the rigorous evaluation of the FVC-ID mechanism, a denser deployment of control centers directly increases the statistical probability that any newly formed network island contains at least one control center. This ensures that fragmented sub-networks can successfully pass the FVC-ID test, thereby satisfying the operational criteria required to maintain localized autonomous control rather than being discarded as non-viable islands.

It is worth noting that this upward trend exhibits clear diminishing marginal returns, leveling off around the 30% threshold. From the perspective of the proposed innovation, this 30% ratio represents an empirical FVC-ID survivability saturation point. Once the number of control centers reaches this threshold, their collective monitoring radii begin to overlap significantly. Adding more control centers beyond this point primarily provides redundant FVC-ID compliance for already-functional islands without substantially mitigating further failures. In practical power grid engineering, control centers are highly complex and expensive hubs. By demonstrating that a moderate 30% deployment ratio is sufficient to satisfy the FVC-ID criteria across the majority of the network, this finding offers a highly practical and cost-effective reference for grid planners. It confirms that maximum survivability can be achieved without the need for economically unrealistic over-provisioning.

C. CPIM-Driven Optimal Placement of Control Centers for Robustness Enhancement

While increasing the capacity redundancy coefficient α and the proportion of control centers generally improves system robustness, such brute-force measures entail substantial infrastructure deployment and maintenance costs. This section further investigates how to maximize system robustness under limited resources by strategically deploying a fixed number of control centers. To this end, we propose the Cohesion-Propagation Importance Metric , a node importance index that integrates local structural stability with global functional reach to identify optimal control center locations. In this experiment, the number of control centers is fixed at 20% of total nodes, with the redundancy coefficient varying in the range $\alpha \in [0, 1]$.

Fig. 5 compares the performance of *CPIM* against three benchmark topological metrics (C_{nc+}, G_+, *MSIS*) and random allocation. Across all scenarios, the *CPIM* strategy consistently yields the highest robustness. Its superiority stems from fundamental alignment with the FVC-ID mechanism. Traditional single-feature metrics inherently fail to guarantee the complex survivability requirements imposed by FVC-ID during cascading failures. For instance, protecting nodes based purely on global betweenness may preserve long-range routes; however, once those routes inevitably break, the resulting fragments often lack local structural redundancy, instantly failing the FVC-ID test and collapsing.

In contrast, *CPIM* mathematically embodies the operational requirements of the CPPS. By integrating local cohesive features with global propagation metrics, it identifies the

network's true structural core. Placing control centers at *CPIM*-identified nodes confers two critical advantages. First, the high local cohesion of these nodes acts as a topological buffer. Even under severe targeted attacks, they anchor tightly-knit local neighborhoods that effortlessly satisfy the FVC-ID mechanism, forming viable, self-sustaining micro-grids. Second, their high global reachability ensures that wide-area shedding and control commands propagate at maximum speed before physical transmission lines breach thermal limits.

Crucially, compared to random placement, *CPIM* achieves significant robustness improvements at lower redundancy coefficients. Given that upgrading physical grid capacity (increasing α) requires years and massive capital investment, mathematically optimizing cyber-resource deployment through *CPIM* offers a software-defined robustness alternative. These findings confirm that the proposed optimization strategy effectively substitutes intelligent cyber-topological planning for expensive physical capacity upgrades, successfully fulfilling the dual objective of minimizing defense costs while maximizing system survivability against targeted cyber-attacks.

V. CONCLUSION

This paper proposes a novel framework for enhancing the robustness of CPPS under targeted cyber attacks. Departing from traditional centralized architectures, we model the communication layer using a distributed control paradigm and introduce the FVC-ID mechanism, which imposes closed-loop operational constraints on surviving Island in both layers to ensure realistic failure assessment. Building on this foundation, we systematically investigate how the deployment of distributed control centers influences system robustness under various coupling patterns. Results reveal that while increasing control redundancy improves survivability, it incurs higher costs. To address this trade-off, we develop a *CPIM*—a topology-based optimization strategy that integrates multiple centrality features to identify optimal control center locations under limited resources. Extensive experiments on benchmark IEEE systems validate that *CPIM* significantly enhances CPPS robustness without excessive redundancy. By bridging complex network theory with practical deployment, this work provides a cost-effective, theoretically grounded solution for strengthening interdependent critical infrastructures against emerging cyber threats.

ACKNOWLEDGMENTS

This research was supported by the National Natural Science Foundation of China under Grants No. 62403174 and 62573170.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the first author upon reasonable request.

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